

Supplementary Material for “Multiplex Depth for Network-valued Data and Applications”

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Introduction

The document serves as a supplement to the main paper and is organized as follows. Section [SI](#) provides definitions for two popularly used network distances. Section [SII](#) shows an efficient algorithm for computing the edge-wise multiplex depth. Section [SIII](#) provides additional simulation results. Section [SIV](#) includes some additional content for the real applications. Theoretical results of gMD for unweighted and weighted networks are illustrated in Section [SV](#). Proofs of the theoretical results are available in Section [SVI](#).

SI Definition

Definition S1 (Fréchet mean). *Let Y be a random element defined on sample space $\mathcal{Y}$ and let $d(\cdot, \cdot)$ be a metric on $\mathcal{Y}$. The set*

$$\psi^m = \arg \inf_{\psi \in \mathcal{Y}} \mathbb{E}_Y [d^2(Y, \psi)] ,$$

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is called the Fréchet mean set of $\mathcal{Y}$.

In this paper, $\mathcal{Y}$ is the network space with N labelled network $\mathcal{G}_{[N]}$. We choose Hamming distance and Frobenius distance as the metric for weighted networks, and only Hamming distance for unweighted networks.

Frobenius distance. The Frobenius distance between G_1 and G_2 is defined as

$$d_{\text{FR}}(G_1, G_2) = \left(\sum_{i \neq j}^N (A_{ij}^1 - A_{ij}^2)^2 \right)^{1/2}.$$

Hamming distance. The Hamming distance between G_1 and $G_2 \in \mathcal{G}_{[N]}$ is defined as

$$d_{\text{HA}}(G_1, G_2) = \sum_{i \neq j}^N \frac{|A_{ij}^1 - A_{ij}^2|}{N(N-1)},$$

where A^1 and A^2 be the matrix representation for G_1 and G_2 .

SII Fast algorithm for EMD

In this section, we design fast algorithms for both weighted and unweighted networks.

SII.1 Fast algorithm for weighted network

In this part, we propose a fast algorithm for calculating $EMD_n^k(G; W)$ for weighted networks when $k \geq 2$, which includes the case of $k = 2$ proposed in [Sun et al. \(2012\)](#). As shown in (9), $EMD_n^k(G; W)$ can be reorganized by

$$\begin{aligned} EMD_n^k(G; W) &= \binom{n}{k}^{-1} \sum_{1 \leq l_1 < l_2 < \dots < l_k \leq n} \sum_{i \neq j} w_{ij} \mathbf{1} \{G \in EM_{H_{i,j}}(G_{l_1}, \dots, G_{l_k})\} \\ &= \binom{n}{k}^{-1} \sum_{i \neq j} w_{ij} \left\{ \binom{n}{k} - \binom{R_{i,j}(G) - 1}{k} - \binom{n - R_{i,j}(G)}{k} \right\}, \end{aligned} \quad (\text{S1})$$

where $R_{i,j}(G)$ is the rank of the edge $e(i, j)$ in network G among all corresponding edges of the networks in $\mathcal{G}_n$, i.e., $\{e_1(i, j), e_2(i, j), \dots, e_n(i, j)\}$. The above formula holds be-

cause there are $\binom{n}{k}$ possible combinations for choosing k networks from $\mathcal{G}_n$, and $G \notin EM_{H_{i,j}}(G_{l_1}, \dots, G_{l_k})$ if and only if the corresponding edges of the selected k networks are all less or all larger than that of G . In this way, the fast algorithm for weighted network is demonstrated in Algorithm S1.

Algorithm S1 Fast algorithm of $EMD_n^k(G; W)$ for weighted network.

Input: An $N \times N \times n$ array $S_{N \times N \times n}$ where each matrix slice $(S_{N \times N, l})$ is the adjacency matrix of a network in $\mathcal{G}_n$; an importance score matrix $W_{N \times N}$; an array $R_{N \times N \times n}$ to store the number of multiplex including the particular edge.

Output: a vector r of length n , representing the $eMD_n^k(G_l; W)$ for $G_l \in \mathcal{G}_n$.

1: Denote $R_{i,j}(G_l)$ as the rank of the edge $e(i, j)$ in network G_l among all corresponding edges of the networks in $\mathcal{G}_n$, i.e., $\{e_1(i, j), e_2(i, j), \dots, e_n(i, j)\}$.

2: **for** $l = 1, 2, \dots, n$ **do**

3: Set

$$D_{i,j,l} = \binom{n}{k} - \binom{R_{i,j}(G_l) - 1}{k} - \binom{n - R_{i,j}(G_l)}{k};$$

4: Set

$$r_l = \binom{n}{k}^{-1} \sum_{i=1}^N \sum_{j=1}^N W_{i,j} D_{i,j,l};$$

5: **return** r .

SII.2 Fast algorithm for unweighted network

In this part, we propose a fast algorithm for calculating $EMD_n^k(G; W)$ for unweighted network when $k \geq 2$. For unweighted network, the binary nature of unweighted network degenerates (S1) to two discrete values. The rough idea is, fixing vertex i and j , the multiplex consisting of k networks does not include network G if and only if there is no edge between vertex i and j in all k networks. We demonstrate the fast algorithm for unweighted networks in Algorithm S2.

Algorithm S2 Fast algorithm of $EMD_n^k(G; W)$ for unweighted network.

Input: An $N \times N \times n$ array $S_{N \times N \times n}$ where each matrix slice $(S_{N \times N, l})$ is the adjacency matrix of a network in $\mathcal{G}_n$; an importance score matrix $W_{N \times N}$; an array $R_{N \times N \times n}$ to store the number of multiplex including the particular edge.

Output: a vector r of length n , representing the $eMD_n^k(G_l; W)$ for $G_l \in \mathcal{G}_n$.

- 1: Denote $p_{i,j} = \#\{k : S_{i,j,k} \neq 0\}$ and $q_{i,j} = n - p_{i,j}$.
- 2: **for** $l = 1, 2, \dots, n$ **do**
- 3: **if** $S_{i,j,k}$ is 0 **then**
- 4: Set $D_{i,j,l} = \binom{n}{k} - \binom{p_{i,j}}{k}$.
- 5: **else**
- 6: Set $D_{i,j,l} = \binom{n}{k} - \binom{q_{i,j}}{k}$;
- 7: Set

$$r_l = \binom{n}{k}^{-1} \sum_{i=1}^N \sum_{j=1}^N W_{i,j} D_{i,j,l};$$

- 8: **return** r .
-

SIII Additional numerical results

SIII.1 Simulations with a large network size

In this section, we perform simulations using the same settings as used in Figures 2 and 4, but with an increased vertex number from $N = 30$ to $N = 200$. The results are presented in Figures S1 and S2. For unweighted networks, as N increases, all methods generally exhibit larger EA. The performance difference between the eMD and Ham remains notable under Scenarios I and III, while becoming relatively smaller under Scenario II. For weighted networks, Ham transitions from divergence ($N = 30$) to convergence ($N = 200$) under Setting II with $r = 0.1$. Increasing N poses a greater challenge for both Ham and Frob under Setting III. However, the performance of eMD remains robust to the choice of N .

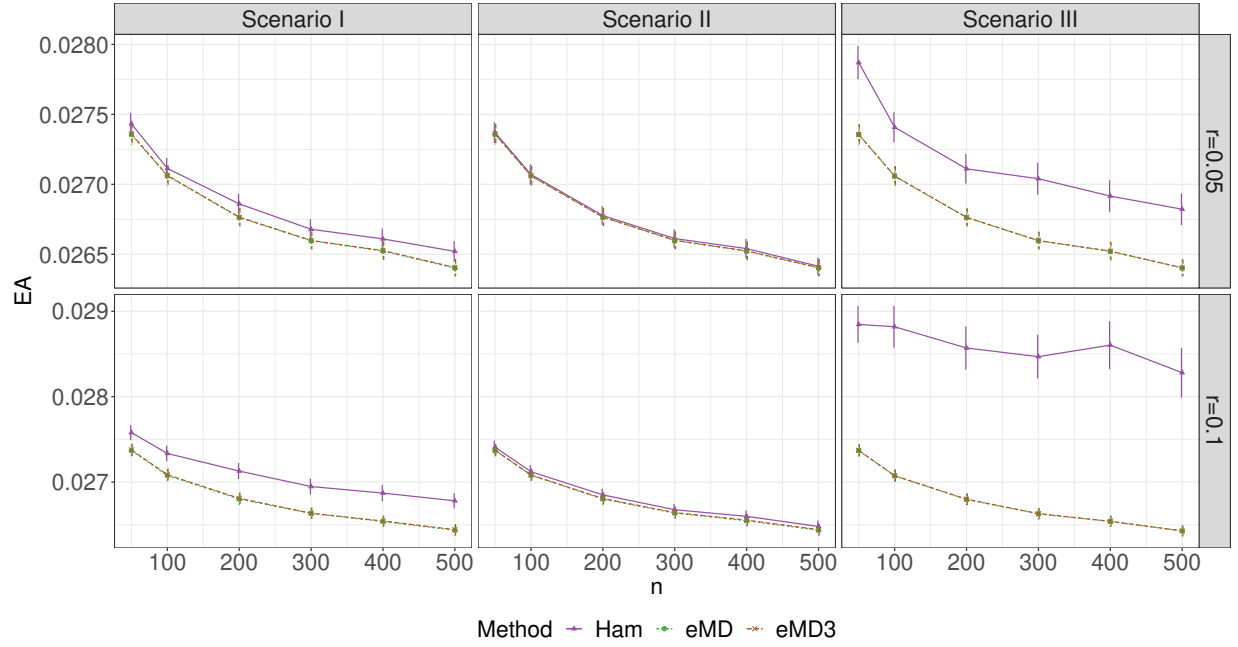
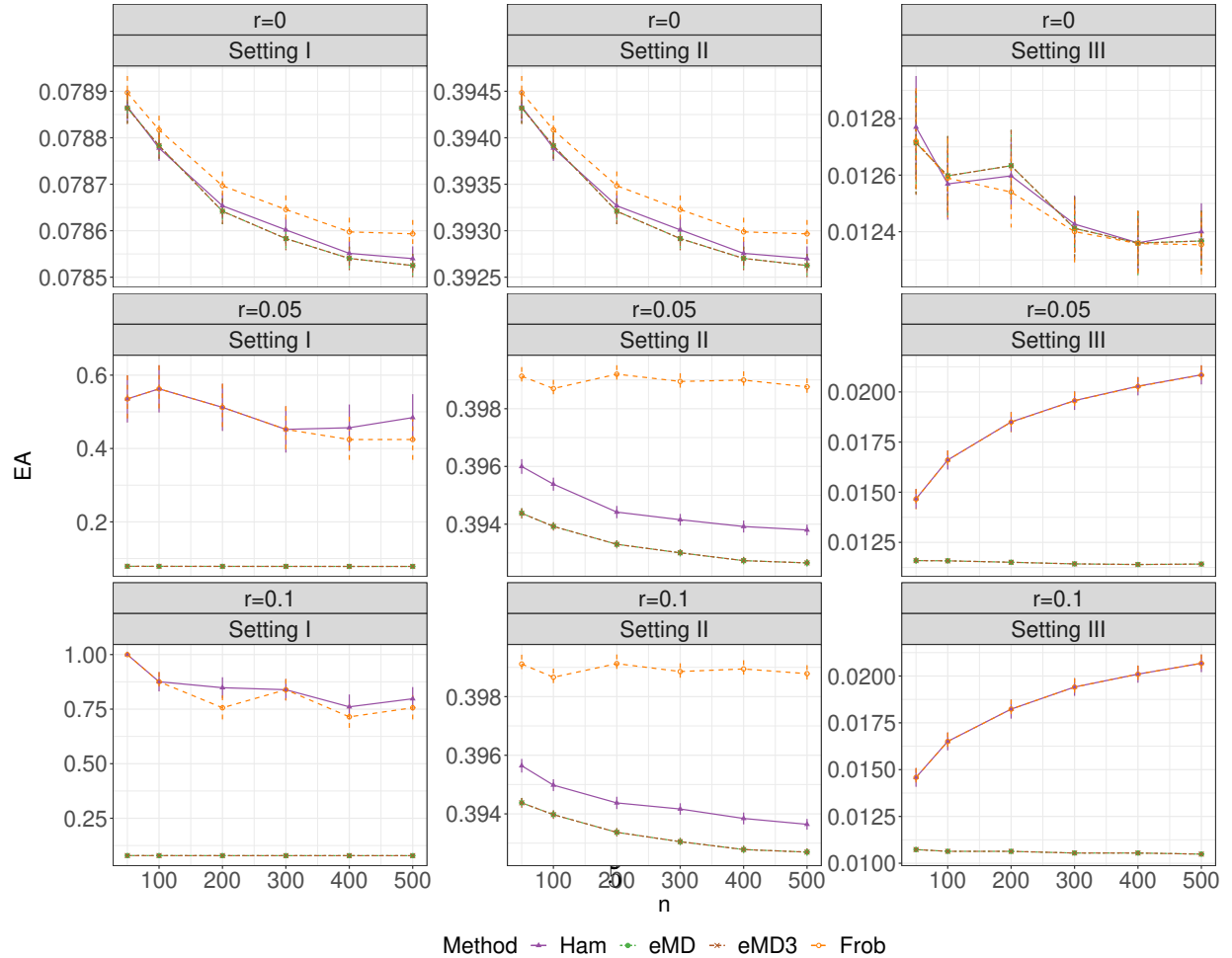


Figure S1: Estimation accuracy (EA) of center estimation for eMD, eMD³, and Ham using equal IS under Scenarios I, II and III. We perform the procedure 200 times for each n (50, 100, 200, ..., 500) and $N = 200$ for each scenario.



SIII.2 Boxplot for unweighted network

In this section, we present the boxplots of EAs under the settings in Figures 2 and 4. As demonstrated in Figure S3, the EAs of eMD and eMD^3 are stabler and lower than those of Ham in the task of central network estimation for unweighted networks. From Figure S4, we notice that the eMD and eMD^3 are quite robust to the outliers, in contrast to the sensitivity of Ham and Frob.

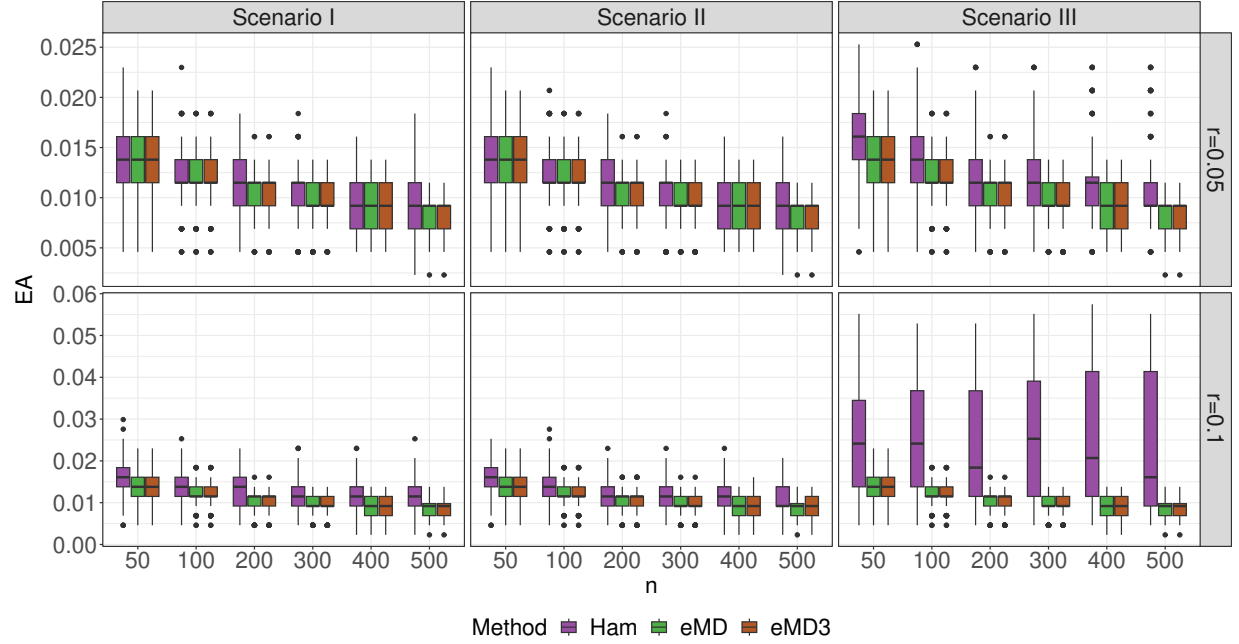


Figure S3: The boxplot of estimation accuracy (EA) of center estimation for eMD, eMD^3 and Ham using equal IS under Scenarios I, II and III. We perform the procedure 200 times for each n (50, 100, 200, ..., 500) and $N = 30$ for each scenario.

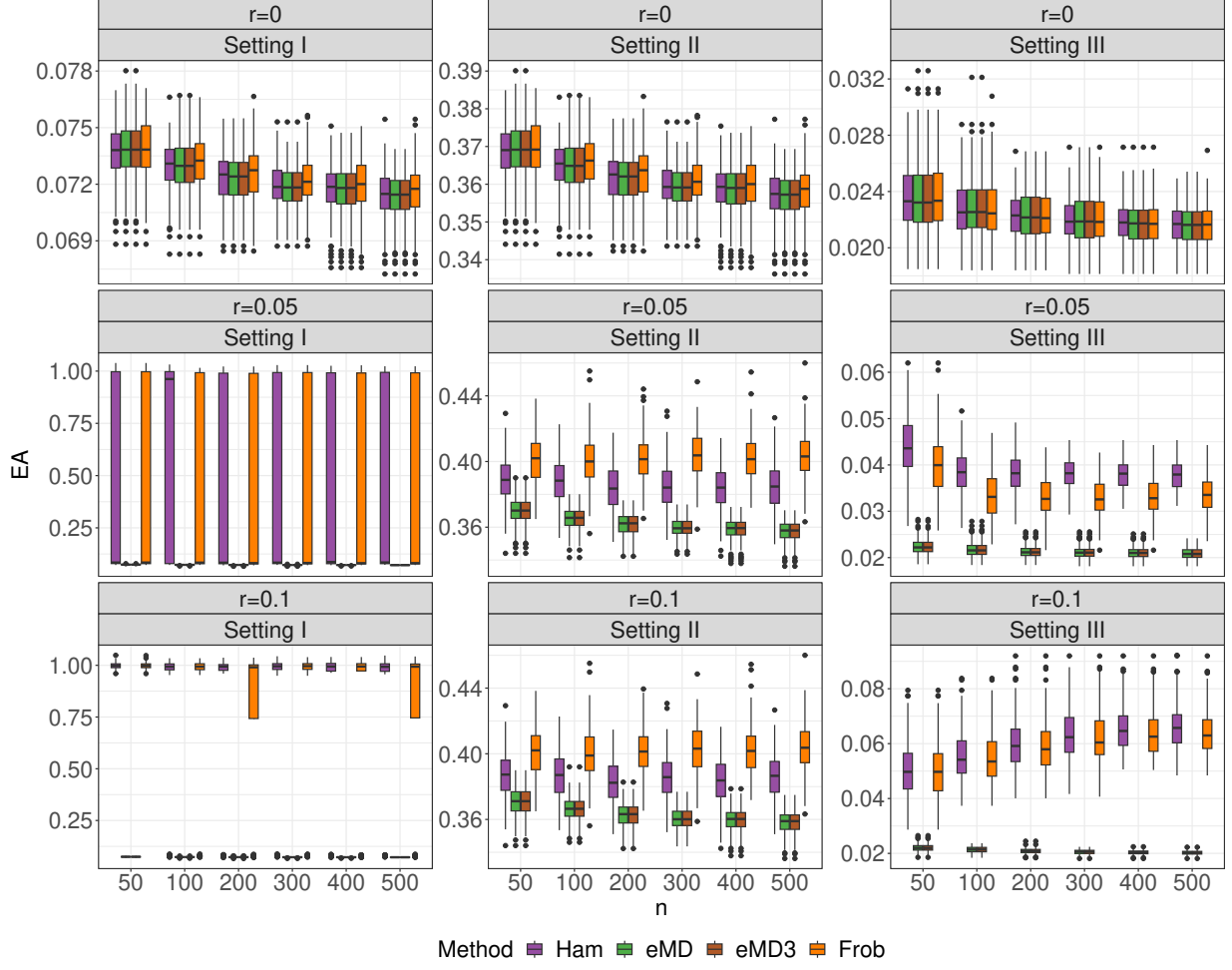


Figure S4: The boxplot of estimation accuracy (EA) of central network estimation for eMD, eMD³, Frob and Ham using equal IS without or with different types of outliers. The first row is the clean model, and the last two rows are contaminated with outlier rates of 5% and 10%, respectively. We fix $N = 30$ and repeat each case 200 times.

SIV Additional real data details

In this section, we provide details for the rank test, the locations of electrodes, and additional results.

Details for the rank test. To assign a rank test based on eMD_n , define the empirical quantile function relative to $\mathcal{G}_n$ as

$$R_n(G, \mathcal{G}_n) = n^{-1} \# \{G_i \in \mathcal{G}_n : eMD_n(G_i) \leq eMD_n(G)\} \in [0, 1],$$

which induces an increasing rank from 1 to n corresponding to $\{R_n(G_i, \mathcal{G}_n) : G_i \in \mathcal{G}_n\}$. We rank according to the indexes of networks if there are ties. Subsequently, we can test whether two network ensembles are from the same population based on the induced rank.

Let $\mathcal{G}_n = \{G_1, G_2, \dots, G_n\}$ and $\mathcal{G}'_m = \{G'_1, G'_2, \dots, G'_m\}$ be two samples of networks from populations $pr_1(\cdot)$ and $pr_2(\cdot)$, respectively. The null hypothesis is $H_0 : pr_1 = pr_2$ and the alternative hypothesis is naturally $H_a : pr_1 \neq pr_2$. We randomly choose a reference network ensemble $\mathcal{G}_{n_0}^0 \sim pr_1(\cdot)$ and calculate the empirical quantile of $\mathcal{G}_n$ and $\mathcal{G}'_m$ relative to $\mathcal{G}_{n_0}^0$, denoted as $\{R_n(G_i, \mathcal{G}_{n_0}^0) : G_i \in \mathcal{G}_n\}$ and $\{R_n(G'_i, \mathcal{G}_{n_0}^0) : G'_i \in \mathcal{G}'_m\}$. Under null hypothesis, we take $\{\mathcal{G}_n, \mathcal{G}'_m\}$ as the reference ensemble. Then, we put the depth together, calculate the increasing rank from 1 to $m+n$ and remove ties similar to the previous procedure. The test statistics is

$$W = \sum_{G'_j \in \mathcal{G}'_m} \text{rank } R(G'_j; \{\mathcal{G}_n, \mathcal{G}'_m\}).$$

Like [Liu and Singh \(1993\)](#), the ranks of $R(G'_j; \{\mathcal{G}_n, \mathcal{G}'_m\})$ behave like sampling n numbers without replacement from $\{1, 2, \dots, n+m\}$ under H_0 . Hence, W has the same distribution as the summation of a sample without replacement from $\{1, 2, \dots, n+m\}$. Smaller W indicates the ranks of $\mathcal{G}'_m$ are lower and hence the depths of them are more shallow on average. Therefore, combining the definition of the alternative hypothesis, we reject H_0 if W is small.

The locations of electrodes are listed in Table [S1](#).

Additional Results. Figure [S5](#) presents the vertex strength heatmaps of two groups with

Table S1: The 61 electrode locations and the corresponding brain regions.

Left	C1 (Left central region), C3 (Left central region), C5 (Left central region), AF1 (Left frontal region), F1 (Left frontal region), F3 (Left frontal region), F5 (Left frontal region), FP1 (Left frontal region), FC1 (Left frontocentral region), FC3 (Left frontocentral region), FC5 (Left frontocentral region), AF7 (Left frontotemporal region), F7 (Left frontotemporal region), O1 (Left occipital region), CP1 (Left parietal region), CP3 (Left parietal region), CP5 (Left parietal region), P1 (Left parietal region), P3 (Left parietal region), P5 (Left parietal region), P7 (Left parietal region), PO1 (Left parietooccipital region), PO7 (Left parietooccipital region), FT7 (Left temporal region), T7 (Left temporal region), TP7 (Left temporooccipital region)
Central	CZ (Central region), FPZ (Frontal central region), FZ (Frontal central region), AFZ (Frontocentral region), FCZ (Frontocentral region), OZ (Occipital central region), CPZ (Parietal central region), PZ (Parietal central region), POZ (Parietooccipital central region)
Right	C2 (Right central region), C4 (Right central region), C6 (Right central region), AF2 (Right frontal region), F2 (Right frontal region), F4 (Right frontal region), F6 (Right frontal region), FP2 (Right frontal region), FC2 (Right frontocentral region), FC4 (Right frontocentral region), FC6 (Right frontocentral region), AF8 (Right frontotemporal region), F8 (Right frontotemporal region), O2 (Right occipital region), CP2 (Right parietal region), CP4 (Right parietal region), CP6 (Right parietal region), P2 (Right parietal region), P4 (Right parietal region), P6 (Right parietal region), P8 (Right parietal region), PO2 (Right parietooccipital region), PO8 (Right parietooccipital region), FT8 (Right temporal region), T8 (Right temporal region), TP8 (Right temporooccipital region)

three maximal eMDs centers. We can observe that the more active regions of the centers in both groups are frontal and parietal, associated with cognitive tasks, which is the same as the findings in Figure 6.

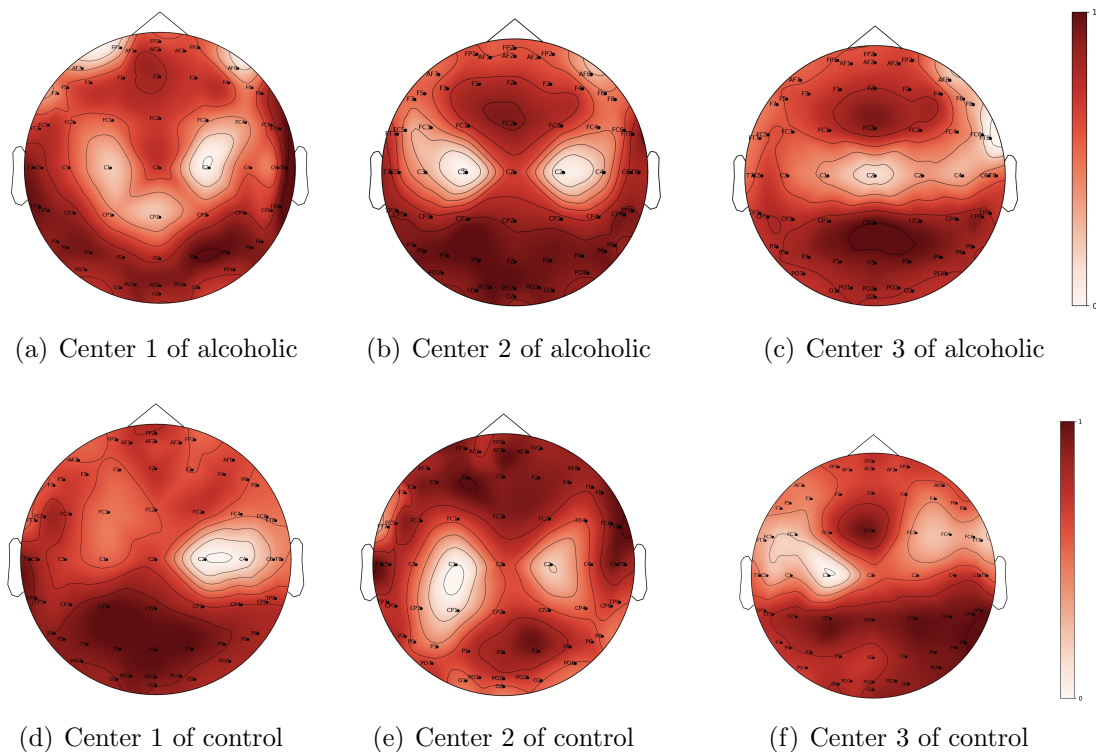


Figure S5: Vertex strength heatmap of subjects with 3 largest eMDs in two groups. We normalized the strength sequence $\mathbf{d}$ to (0,1), where the strength sequence $\mathbf{d} = (d_1, \dots, d_N)$ for each subject.

SV Theoretical results about gMD

In this section, we examine the depth property and convergence of gMD.

SV.1 Depth property for unweighted networks

We first establish the depth property for gMD for unweighted networks. The depth property of eMD for unweighted networks can be naturally established in the same way, as eMD can be viewed as a summation of gMD on $N \times N$ networks. It is necessary to introduce an assumption and a definition in advance.

Assumption S1 (Independence). *Suppose $(\mathcal{G}_{[N]}, \mathcal{F}, pr_{G^c})$ is the probability space for unweighted networks equipped with a center $G^c = (V, E^c)$ and $p_{ij}^{(per)} = pr(A_{ij} \neq A_{ij}^c)$, the marginal probability of perturbing from the center. Suppose $p_{ij}^{(per)} < \frac{1}{2}$ and the perturbations are independently distributed.*

Definition S2 (Transition path). We denote the start network and the end network as $G^{start} = (V, E^{start})$ and $G^{end} = (V, E^{end})$, respectively. We also let N_0 be the cardinality of the set $\{(i, j) : A_{ij}^{start} \neq A_{ij}^{end}\}$. We say $\text{Pa}(G^{start}, G^{end}) = (G_0, G_1, \dots, G_{N_0})$ is a transition path from G^{start} to G^{end} if (i) $G_0 = G^{start}$, $G_{N_0} = G^{end}$; (ii) $\#\{(i, j) : A_{ij}^k \neq A_{ij}^{k+1}\} = 1$, $k = 0, 1, \dots, N_0 - 1$.

Property S1. Consider a given path $\text{Pa}(G^{start}, G^{end}) = (G_0, G_1, \dots, G_{N_0})$. (i) There are $N_0!$ distinguished transition paths; (ii) $\#\{(i, j) : A_{ij}^{start} \neq A_{ij}^k\} = k$, $k = 0, 1, \dots, N_0$;

Property S2. Consider a given path $\text{Pa}(G^{start}, G^{end}) = (G_0, G_1, \dots, G_{N_0})$. (i) Any two networks on the path are distinct. (ii) The sets of flipped edges are nested. Specifically, denote $D_{edge}(G_a, G_b) = \{(i, j) : A_{ij}^a \neq A_{ij}^b\}$, then

$$D_{edge}(G_k, G^{start}) \subset D_{edge}(G_l, G^{start}) \quad \text{and} \quad D_{edge}(G_l, G^{end}) \subset D_{edge}(G_k, G^{end}),$$

for all $0 \leq k < l \leq N_0$.

Assumption S1 is satisfied by various network-valued data models, such as the Erdős-Rényi model (Gilbert, 1959), stochastic block model (Wang and Wong, 1987; Zhang et al., 2020), the latent space model (Hoff et al., 2002; Ma et al., 2020), unimodal network distributions (Lunagómez et al., 2021), and a collection of noisy networks (Chang et al., 2020). Properties S1 and S2 imply two natural properties. First, identical edges between the start and end networks will not flip in any transition path. Second, the transition paths are nested, so that replacing an arbitrary segment of a long path with another creates a new long path.

Theorem S1 (Depth property of gMD for unweighted networks). Under Assumption S1, we have

(a) *Maximality at the center.* $gMD_{pr}^K(\cdot)$ is uniquely maximized at G^c ;

(b) *Monotonicity relative to the center.* Consider $G \in \mathcal{G}_{[N]}$ and $\text{Pa}(G^c, G) = (G_0, G_1, \dots, G_{N_0})$.

For any $0 \leq k < l \leq N_0$, $gMD_{pr}^K(G_l) < gMD_{pr}^K(G_k)$;

(c) *Vanishing at infinity.* $\sup_{\|A^G\|_\infty \geq M} gMD_{pr}^K(G) \rightarrow 0$ and $\sup_{\|A^G\|_\infty \geq M} gMD_n^K(G) \xrightarrow{a.s.} 0$, as $M \rightarrow \infty$;

(d) *Affine invariance.*

(d)-(i) *Given an arbitrary unweighted network $G^T = (V, E^T)$, consider $T : \mathcal{G}_{[N]} \mapsto \mathcal{G}_{[N]}$ where $T(G) = (V, \Delta(E, E^T))^1$, for any $G = (V, E) \in \mathcal{G}_{[N]}$, then $gMD_n^K(G) = gMD_n^{K, T(G_n)}(T(G))$ and $gMD_{pr}^K(G) = gMD_{pr \circ T^{-1}}^K(T(G))$;*

(d)-(ii) *Consider $\pi : [N] \mapsto [N]$, the permutation transformation of vertices, which induces a transformation $\pi^G : \mathcal{G}_{[N]} \mapsto \mathcal{G}_{[N]}$. Then $gMD_n^K(G) = gMD_n^{K, \pi^G(G_n)}(\pi^G(G))$ and $gMD_{pr}^K(G) = gMD_{pr \circ (\pi^G)^{-1}}^K(\pi^G(G))$.*

Theorem S1 analyzes the four depth properties of the gMD for unweighted networks. If the perturbation probability is homogeneous, that is, $p_{ij}^{(per)} = p$ for some $p < 1/2$, gMD strictly decreases with the hamming distance to the central network. We can then eliminate the concept of the transition path. The depth properties of eMD for unweighted networks are similar to Theorem S1 except that the independence assumption can be relaxed due to the additivity of expectation; we therefore omit these. Finally, the transformation T induced by G^T in Theorem S1(d)-(i) is equal to applying an exclusive OR on the corresponding adjacent matrices, which is the affine transformation adapted to the unweighted network. More intuitive, for each $e(i, j) \in E^T$, we remove $e(i, j)$ from E if $e(i, j) \in E$, or add $e(i, j)$ into E if $e(i, j) \notin E$.

SV.2 Depth property for weighted networks

In this section, we develop the depth properties of gMD for weighted networks. Consider the probability space of weighted networks $(\mathcal{G}_{[N]}, \mathcal{F}, pr)$. The marginal distribution of the edge weight is generally required to be absolutely continuous, tight, and symmetric to the central network. The depth properties (a)-(c) of gMD for weighted networks are extended from the modified band depth for functional data (López-Pintado and Romo, 2009).

Theorem S2 (Depth property of gMD for weighted networks). *Under Assumptions 1 and 2, we have*

(a) *Maximality at the center. If the density f is further positive in a neighborhood of the center of G^c , then $gMD_{pr}^K(\cdot)$ is uniquely maximized at G^c ;*

¹ $\Delta(A, B)$ is the subtraction of A and B .

(b) *Monotonicity relative to the center.* Given $G \in \mathcal{G}_{[N]}$ and a scaling parameter $\alpha > 0$, define G^α as a network whose adjacency matrix is $\alpha(A^G - A^c)$, then $gMD_{pr}^K(\alpha(G - G^c))$ is a monotone nonincreasing function with respect to α ;

(c) *Vanishing at infinity.* $\sup_{\|A^G\|_\infty \geq M} gMD_{pr}^K(G) \rightarrow 0$ and $\sup_{\|A^G\|_\infty \geq M} gMD_n^K(G) \xrightarrow{a.s.} 0$, as $M \rightarrow \infty$.

(d) *Affine Invariance.*

(d)-(i) Consider $T : \mathcal{G}_{[N]} \mapsto \mathcal{G}_{[N]}$ where the adjacency matrix of G and $T(G)$ are A and $C \odot A + D$ with $C, D \in \mathbb{R}^{N \times N}$, respectively. If $C_{ij} \neq 0$, then $gMD_n^K(G) = gMD_n^{K, T(\mathcal{G}_n)}(T(G))$ and $gMD_{pr}^K(G) = gMD_{pr \circ T^{-1}}^K(T(G))$;

(d)-(ii) Consider $\pi : [N] \mapsto [N]$, the permutation transformation of vertices, which induces a transformation $\pi^G : \mathcal{G}_{[N]} \mapsto \mathcal{G}_{[N]}$. Then, $gMD_n^K(G) = gMD_n^{K, \pi^G(\mathcal{G}_n)}(\pi^G(G))$ and $gMD_{pr}^K(G) = gMD_{pr \circ (\pi^G)^{-1}}^K(\pi^G(G))$.

SV.3 Convergence

In this section, we introduce the pointwise and uniform convergence of gMD, as well as the consistency of center estimated by gMD. In addition, we show gMD and eMD is continuous for weighted network.

To establish the convergence of the empirical and the population deepest networks, the metrics of the networks are required. [Donnat and Holmes \(2018\)](#) lists the most common metrics of $d(\cdot, \cdot) : \mathcal{G}_{[N]} \times \mathcal{G}_{[N]} \rightarrow \mathbb{R}$ for networks. We note that once N is fixed, the sample space for the unweighted network is finite. Thus, there exists $\delta_N > 0$ small enough such that $d(G_a, G_b) > \delta_N$, for arbitrary $G_a, G_b \in \mathcal{G}_{[N]}$ that $G_a \neq G_b$. Recall that in our setting, $0 \leq p_{ij}^{(per)} < 1/2$ and hence $gMD_{pr}^K(\cdot)$ is uniquely maximized at G^c . There exists a uniform $\epsilon_N > 0$ small enough such that $gMD_{pr}^K(G^c) - gMD_{pr}^K(G) > \epsilon_N$, for all $G \in \mathcal{G}_{[N]}$ such that $G \neq G^c$ attributing to the finite sample space. From the above discussion, $d(G^c, G) > \delta_N$ implies $gMD_{pr}^K(G^c) - gMD_{pr}^K(G) > \epsilon_N$ and vice versa. This property is essential for the proof of Theorem [S3](#). For weighted networks, Theorem [S3](#) is a direct corollary of Theorem 2 in [López-Pintado and Romo \(2009\)](#).

Theorem S3 (gMD for unweighted and weighted networks). *Suppose $G_1, G_2, \dots, G_n$ are n*

samples independently generated from a probability distribution for unweighted or weighted networks. Then,

- (a) $\sup_{G \in \mathcal{G}_{[N]}} |gMD_n^K(G) - gMD_{pr}^K(G)| \xrightarrow{a.s.} 0$, as $n \rightarrow \infty$;
- (b) If $gMD_{pr}^K(\cdot)$ is uniquely maximized at G^c and set $\tilde{\mathbf{G}}_n = \arg \max_{G \in \mathcal{G}_{[N]}} gMD_n^K(G)$, then $\tilde{\mathbf{G}}_n \xrightarrow{a.s.} G^c$, as $n \rightarrow \infty$;
- (c) If $gMD_{pr}^K(\cdot)$ is uniquely maximized at G^c and set $\hat{\mathbf{G}}_n = \arg \max_{G \in \{\mathbf{G}_1, \mathbf{G}_2, \dots, \mathbf{G}_n\}} gMD_n^K(G)$, then $\hat{\mathbf{G}}_n \xrightarrow{a.s.} G^c$, as $n \rightarrow \infty$.

Proposition S1 (Continuous for weighted networks). *For weighted networks, (a) $gMD_{pr}^K(\cdot)$ is upper-semicontinuous. If Assumption 1 is satisfied, then $gMD_{pr}^K(\cdot)$ is continuous.*

(b) Given $W \in \mathbb{R}^{n \times n}$, $eMD_{pr}^K(\cdot; W)$ is upper-semicontinuous. If Assumption 1 is satisfied, then $eMD_{pr}^K(\cdot; W)$ is continuous.

Proof of Proposition S1. (a) is a direct corollary of Theorem 3(4) in López-Pintado and Romo (2006).

(b) is trivial once we notice that $eMD_{pr}^K(\cdot, W)$ is a finite summation of $gMD_{pr}^K(\cdot)$ for networks restricted to $\{V = \{i, j\}, E = \{e(i, j)\}\}$ because the summation of finite upper-semicontinuous/continuous functions is still an upper-semicontinuous/continuous function. $\square$

SVI Proofs of theoretical results

This section proves the theoretical results. As eMD is a weighted average of gMD restricted to each edge, we firstly justify the theory about gMD and then fill in the necessary details for proving eMD.

SVI.1 Proof of Theorems S1 to S3

We first prove Properties S1 and S2.

Proof of Property S1. (i) Since there are N_0 discrepant edges between the start network and the end network, there are N_0 possible choices for G_1 . Once the first network is determined,

there are $N_0 - 1$ discrepant edges between G_1 and the end network. By induction, there are $N_0!$ possible paths.

(ii) Since $\# \{(i, j) : A_{ij}^k \neq A_{ij}^{k+1}\} = 1, k = 0, 1, \dots, N_0 - 1$ by the definition of transition path, we get $\# \{(i, j) : A_{ij}^{k+1} \neq A_{ij}^{start}\} = \# \{(i, j) : A_{ij}^k \neq A_{ij}^{start}\} \pm 1, k = 0, 1, \dots, N_0 - 1$. Together with $\# \{(i, j) : A_{ij}^{start} \neq A_{ij}^{end}\} = N_0$, the sign must be “plus”, i.e.,

$$\# \{(i, j) : A_{ij}^{k+1} \neq A_{ij}^{start}\} = \# \{(i, j) : A_{ij}^k \neq A_{ij}^{start}\} + 1, \quad (\text{S2})$$

$k = 0, 1, \dots, N_0$. Hence, we get the desired result. $\square$

Proof of Property S2. (i) Without loss of generality, if $G_k = G^{start}$ for some $k > 0$, then $\# \{(i, j) : A_{ij}^k \neq A_{ij}^{end}\} = N_0$. However, according to (S2), at most $N_0 - k < N_0$ edges jump from G_k to G^{end} , which causes contradiction.

(ii) Without loss of generality, we consider two neighboring networks. Assume that $D_{edge}(G_k, G_0) \not\subset D_{edge}(G_{k+1}, G_0)$ for some $k \geq 0$. Then, there exists (i_0, j_0) such that $A_{i_0 j_0}^{start} = A_{i_0 j_0}^{k+1} \neq A_{i_0 j_0}^k$. According to Property S1 and (S2), we have

$$\begin{aligned} \# \{(i, j) : A_{ij}^{start} \neq A_{ij}^{k+1}\} &= \# \{ \{(i, j) : A_{ij}^{start} \neq A_{ij}^k\} \setminus \{(i, j) : A_{ij}^k \neq A_{ij}^{k+1}\} \} \\ &= \# \{ \{(i, j) : A_{ij}^{start} \neq A_{ij}^k\} \setminus \{(i_0, j_0)\} \} \\ &= k - 1 \end{aligned}$$

which contradicts to Property S1. $\square$

Now, we can establish the proof of Theorem S1.

Proof of Theorem S1. We prove this theorem in the order (c), (b), (a) and (d).

(c) *Vanishing at infinity.* Vanishing at infinity naturally holds for unweighted networks with fixed vertices because the norm of the adjacency matrix is always finite. Therefore, $\sup_{\|A^G\|_\infty \geq M} gMD_{pr}^K(G) = \sup_{\|A^G\|_\infty \geq M} gMD_n^K(G) = 0$ for arbitrary $G \in \mathcal{G}_{[N]}$ once we take $M = N^2 + 1$.

(b) *Monotonicity relative to the center.* For simplicity, let $K = 2$ and we only have to prove that $gMD_{pr}^K(G_k) \geq gMD_{pr}^K(G_{k+1})$, where G_k and G_{k+1} are arbitrary two neighboring

networks on the transition path $\text{Pa}(G^c, G)$. The formula can be further simplified as

$$\text{pr}\{G_k \in M(\mathbf{G}_a, \mathbf{G}_b)\} \geq \text{pr}\{G_{k+1} \in M(\mathbf{G}_a, \mathbf{G}_b)\},$$

where $\mathbf{G}_a, \mathbf{G}_b \sim \text{pr}(\cdot)$ are independent. Since only one edge flip between the neighboring networks, define (i_0, j_0) such that $0 = A_{i_0 j_0}^c = A_{i_0 j_0}^0 = A_{i_0 j_0}^k \neq A_{i_0 j_0}^{k+1} = A_{i_0 j_0}^{N_0} = 1$. Because other edges of G_k and G_{k+1} are identical, the following discussion is based on conditioning on other edges. Moreover, under Assumption S1, we only need to prove

$$\text{pr}\{A_{i_0 j_0}^k \in [\mathbf{A}_{i_0 j_0}^{\text{down}}, \mathbf{A}_{i_0 j_0}^{\text{up}}]\} \geq \text{pr}\{A_{i_0 j_0}^{k+1} \in [\mathbf{A}_{i_0 j_0}^{\text{down}}, \mathbf{A}_{i_0 j_0}^{\text{up}}]\},$$

where $\mathbf{A}_{i_0 j_0}^{\text{up}} = \mathbf{A}_{i_0 j_0}^a \vee \mathbf{A}_{i_0 j_0}^b$ and $\mathbf{A}_{i_0 j_0}^{\text{down}} = \mathbf{A}_{i_0 j_0}^a \wedge \mathbf{A}_{i_0 j_0}^b$. Now,

$$\begin{aligned} \text{pr}\{A_{i_0 j_0}^k \in [\mathbf{A}_{i_0 j_0}^{\text{down}}, \mathbf{A}_{i_0 j_0}^{\text{up}}]\} &= 1 - \text{pr}\{[\mathbf{A}_{i_0 j_0}^a = 1] \cap [\mathbf{A}_{i_0 j_0}^b = 1]\} = 1 - \left(p_{i_0 j_0}^{(\text{per})}\right)^2, \\ \text{pr}\{A_{i_0 j_0}^{k+1} \in [\mathbf{A}_{i_0 j_0}^{\text{down}}, \mathbf{A}_{i_0 j_0}^{\text{up}}]\} &= 1 - \text{pr}\{[\mathbf{A}_{i_0 j_0}^a = 0] \cap [\mathbf{A}_{i_0 j_0}^b = 0]\} = 1 - \left(1 - p_{i_0 j_0}^{(\text{per})}\right)^2. \end{aligned}$$

Consequently, we can conclude (b) is true because $p_{i_0 j_0}^{(\text{per})} < 1/2$.

(a) *Maximization at center.* For any $G \in \mathcal{G}_{[N]}$, we can construct a transition path from G^c to G . Then the maximization at center immediately holds according to (b).

(d) *Affine invariance.* Theorem S1(d,ii) stands because conducting permutation transformation of vertices can be viewed as changing the label of vertex and the gMD does not depend on the label of vertices by definition. To show (d,i), we only have to show that for $G_1, \dots, G_k \in \mathcal{G}_n$,

$$\mathbf{1}\{G \in M(G_1, G_2, \dots, G_k)\} = \mathbf{1}\{T(G) \in M(T(G_1), T(G_2), \dots, T(G_k))\},$$

which equals to prove

$$\wedge_{i,j} \mathbf{1}\{G \in EM_{H_{i,j}}(G_1, G_2, \dots, G_k)\} = \wedge_{i,j} \mathbf{1}\{T(G) \in EM_{H_{i,j}}(T(G_1), T(G_2), \dots, T(G_k))\}.$$

The problem is further simplified to show

$$\mathbf{1} \{G \in EM_{H_{i_0, j_0}}(G_1, G_2, \dots, G_k)\} = \mathbf{1} \{T(G) \in EM_{H_{i_0, j_0}}(T(G_1), T(G_2), \dots, T(G_k))\} \quad (\text{S3})$$

for any fixed edge (i_0, j_0) . Recall that A^T is the adjacency matrix of G^T . There are eight cases splitting by (i) $A_{i_0 j_0} = \pm 1$; (ii) $A_{i_0 j_0}^T = \pm 1$; (iii) whether $G \in EM_{H_{i_0, j_0}}(G_1, G_2, \dots, G_k)$.

It's enough to consider the case when (i), (ii) and (iii) are positive, i.e. $A_{i_0 j_0} = 1$, $A_{i_0 j_0}^T = 1$ and $G \in EM_{H_{i_0, j_0}}(G_1, G_2, \dots, G_k)$ and the rests seven cases are similar. From condition (i) and (iii), there exists $l \in \{1, 2, \dots, k\}$ such that $A_{i_0 j_0}^l = 1$. The adjacency matrices of $T(G)$ and $T(G_k)$ are $A_{i_0 j_0}^{T(G)} = 0$ and $A_{i_0 j_0}^{T(G_k)} = 0$ according to the definition of the transformation T . Then, (S3) stands. After traversing all edges, the equivalence of multiplex depth function can be derived directly from the expression (sample version) or from the transformation of random variable (population version). $\square$

Proof of Theorem S2. (a), (b) and (c) are Corollary of Theorem 1(1)-1(3) in López-Pintado and Romo (2009), respectively. In addition, Theorem S2(d,ii) stands because conducting permutation transformation of vertices can be viewed as changing the label of vertex and the eMD does not depend on the label of vertices by definition. Therefore, we just consider (d,i) network affine invariance here. Recall the basic component of gMD in (3) is

$$\begin{aligned} M(G_1, G_2, \dots, G_K) &= \{G : G^{down} \preceq G \preceq G^{up}\} \\ &= \{(V, E) : e^{down}(i, j) \leq e(i, j) \leq e^{up}(i, j), i, j = 1, 2, \dots, N, i \neq j\} \\ &= \{G : A^{down} \preceq A^G \preceq A^{up}\}, \end{aligned}$$

where A^{down} , A^{up} and A^G are the adjacency matrices of G^{down} , G^{up} and G and $\preceq$ for matrices is pointwise comparison. Denote $(C \odot A + D)_{ij}^{down} = \wedge_{k=1}^K (C \odot A^k + D)_{ij} = \wedge_{k=1}^K (C_{ij} A_{ij}^k + D_{ij})$

and $(C \odot A + D)^{up}$ similarly, then

$$\begin{aligned}
& \{G : (C \odot A + D)^{down} \preceq C \odot A^G + D \preceq (C \odot A + D)^{up}\} \\
& \stackrel{(i)}{=} \{G : (C \odot A)^{down} \preceq C \odot A^G \preceq (C \odot A)^{up}\} \\
& \stackrel{(ii)}{=} \cap_{k,l} \{G : (C_{ij}A_{ij})^{down} \preceq C_{ij}A_{ij}^G \preceq (C_{ij}A_{ij})^{up}\} \\
& \stackrel{(iii)}{=} \left\{ \cap_{k,l:C_{ij}>0} \{G : A_{ij}^{down} \preceq A_{ij}^G \preceq A_{ij}^{up}\} \cap \left\{ \cap_{k,l:C_{ij}<0} \{G : (C_{ij}A_{ij})^{down} \preceq C_{ij}A_{ij}^G \preceq (C_{ij}A_{ij})^{up}\} \right\} \right\} \\
& \stackrel{(iv)}{=} \left\{ \cap_{k,l:C_{ij}>0} \{G : A_{ij}^{down} \preceq A_{ij}^G \preceq A_{ij}^{up}\} \cap \cap_{k,l:C_{ij}<0} \{G : -(C_{ij}A_{ij})^{up} \preceq -C_{ij}A_{ij}^G \preceq -(C_{ij}A_{ij})^{down}\} \right\} \\
& \stackrel{(v)}{=} \left\{ \cap_{k,l:C_{ij}>0} \{G : A_{ij}^{down} \preceq A_{ij}^G \preceq A_{ij}^{up}\} \cap \cap_{k,l:C_{ij}<0} \{G : (-C_{ij}A_{ij})^{down} \preceq -C_{ij}A_{ij}^G \preceq (-C_{ij}A_{ij})^{up}\} \right\} \\
& \stackrel{(vi)}{=} \left\{ \cap_{k,l:C_{ij}>0} \{G : A_{ij}^{down} \preceq A_{ij}^G \preceq A_{ij}^{up}\} \cap \cap_{k,l:C_{ij}<0} \{G : A_{ij}^{down} \preceq A_{ij}^G \preceq A_{ij}^{up}\} \right\} \\
& \stackrel{(vii)}{=} \{G : A^{down} \preceq A^G \preceq A^{up}\}.
\end{aligned}$$

To prove the above equation, we use the fact (i) adding a constant and multiplying a positive value doesn't change the order; (ii) multiplying a negative value reverses the order. More detailed, for (i), adding a constant doesn't change the order; (ii) holds according to the definition of $\preceq$; For (iii), We divide the edges into two groups according to whether C_{ij} is greater than zero and use the fact that multiplying a positive value doesn't change the order; (iv) stands because multiplying a negative value reverses the order; (v) can be obtained by noticing that $-(C_{ij}A_{ij})^{up} = -\vee_{k=1}^K C_{ij}A_{ij}^k = \wedge_{k=1}^K (-C_{ij}A_{ij}^k) = (-C_{ij}A_{ij})^{down}$ and $-(C_{ij}A_{ij})^{down} = -\wedge_{k=1}^K C_{ij}A_{ij}^k = \vee_{k=1}^K (-C_{ij}A_{ij}^k) = (-C_{ij}A_{ij})^{up}$; (vi) again uses the fact that multiplying a positive value doesn't change the order; (vii) merges the two groups of edges together and applies the definition of $\preceq$. After that, the equivalence of gMD can be derived directly from the expression (empirical version) or from the transformation of random variable (population version). $\square$

Proof of Theorem S3. For weighted network, this is a corollary of Theorem 2 in [López-Pintado and Romo \(2009\)](#).

Now, let's focus on unweighted networks. For (a), it is straightforward to show that

$$\begin{aligned} \sup_{G \in \mathcal{G}_{[N]}} |gMD_n^K(G) - gMD_{pr}^K(G)| &\leq \sum_{G \in \mathcal{G}_{[N]}} |gMD_n^K(G) - gMD_{pr}^K(G)| \\ &\xrightarrow{a.s.} 0, \end{aligned}$$

where the convergence is it is the summation of $2^{N \times N}$ terms (finite terms) and each term converges according to Proposition 1. Hence, we have $gMD_n^K(G) \xrightarrow{a.s.} gMD_{pr}^K(G)$.

(b) The weak consistency is immediately implied by the property of M estimator (Theorem 5.7 in [van der Vaart \(2000\)](#)). As for the strong consistency, since the sample space $\mathcal{G}_{[N]}$ is finite, we have

$$\begin{aligned} &\tilde{\mathbf{G}}_n \xrightarrow{a.s.} G^c \\ \Leftrightarrow &pr \left[\limsup_n \sup_{l \geq n} \left\{ d(\tilde{\mathbf{G}}_l, G^c) > \delta \right\} \right] = 0, \quad \forall \delta \text{ such that } \delta_N > \delta > 0 \\ \Leftrightarrow &pr \left[\limsup_n \sup_{l \geq n} \left\{ gMD_{pr}^K(G^c) - gMD_{pr}^K(\tilde{\mathbf{G}}_l) > \epsilon \right\} \right] = 0, \quad \forall \epsilon \text{ such that } \epsilon_N > \epsilon > 0 \\ \Leftrightarrow & \\ pr \left[\limsup_n \sup_{l \geq n} \left\{ gMD_{pr}^K(G^c) - gMD_l^K(G^c) + gMD_l^K(G^c) - gMD_l^K(\tilde{\mathbf{G}}_l) + gMD_l^K(\tilde{\mathbf{G}}_l) - gMD_{pr}^K(\tilde{\mathbf{G}}_l) > \epsilon \right\} \right] & \\ = 0 & \\ \Leftrightarrow & \\ pr \left[\limsup_n \sup_{l \geq n} \left\{ gMD_{pr}^K(G^c) - gMD_l^K(G^c) + gMD_l^K(\tilde{\mathbf{G}}_l) - gMD_{pr}^K(\tilde{\mathbf{G}}_l) > \epsilon + gMD_l^K(\tilde{\mathbf{G}}_l) - gMD_l^K(G^c) \right\} \right] & \\ = 0 & \\ \Leftarrow pr \left[\limsup_n \sup_{l \geq n} \left\{ gMD_{pr}^K(G^c) - gMD_l^K(G^c) + gMD_l^K(\tilde{\mathbf{G}}_l) - gMD_{pr}^K(\tilde{\mathbf{G}}_l) > \epsilon \right\} \right] &= 0 \\ \Leftarrow pr \left[\limsup_n \sup_{l \geq n} \left\{ gMD_{pr}^K(G^c) - gMD_l^K(G^c) > \epsilon \right\} \right] + pr \left[\limsup_n \sup_{l \geq n} \left\{ gMD_l^K(\tilde{\mathbf{G}}_l) - gMD_{pr}^K(\tilde{\mathbf{G}}_l) > \epsilon \right\} \right] &= 0, \end{aligned}$$

where the first three lines holds because of the discussion in Section 3.2; the connection between the third last and the second last line is because $gMD_l^K(\tilde{\mathbf{G}}_l) - gMD_l^K(G^c) \geq 0$ by the definition of $\tilde{\mathbf{G}}_l$; and the last line is verified by (a).

(c) Similar to (b), to prove $\widehat{\mathbf{G}}_n \xrightarrow{a.s.} G^c$, we just need to show

$$pr \left[\limsup_n \sup_{l \geq n} \left\{ gMD_{pr}^K(G^c) - gMD_l^K(G^c) + gMD_l^K(\widehat{\mathbf{G}}_l) - gMD_{pr}^K(\widehat{\mathbf{G}}_l) > \epsilon + gMD_l^K(\widehat{\mathbf{G}}_l) - gMD_l^K(G^c) \right\} \right]$$

equals to zero. In fact, $G^c \in \mathcal{G}_n$ can imply $gMD_l^K(\widehat{\mathbf{G}}_l) - gMD_l^K(G^c) \geq 0$. Therefore, we only have to prove $pr \left[\lim_n \sup_{l \geq n} \{G^c \in \mathcal{G}_n\} \right] = 1$, which is equivalent to

$$pr [G^c \notin \{\mathbf{G}_1, \mathbf{G}_2, \dots, \mathbf{G}_n\}] \leq (1 - pr(\mathbf{G} = G^c))^n \rightarrow 0$$

because the event is monotonic. The remaining part is similar to (b). □

SVI.2 Proof of Theorems 1 and 2

The generalization from gMD to eMD for weighted networks is similar to the approach for multivariate data that is proposed in [Claeskens et al. \(2014\)](#).

Proof of Theorem 1. Recall the relationship between EMD_n^k and MD_n^k for weighted networks:

$$EMD_n^k(G; W) = \sum_{i \neq j} w_{ij} \left(\binom{n}{k}^{-1} \sum_{1 \leq l_1 < l_2 < \dots < l_k \leq n} \mathbf{1}\{G \in EM_{H_{i,j}}(G_{l_1}, G_{l_2}, \dots, G_{l_k})\} \right)$$

and

$$EMD_{pr}^k(G; W) = \sum_{i \neq j} w_{ij} pr\{G \in EM_{H_{i,j}}(\mathbf{G}_1, \mathbf{G}_2, \dots, \mathbf{G}_k)\}.$$

In addition, the center G^c is defined according to the marginal distribution of edges and hence the G^c in Theorem [S2](#) and Theorem 1 are the same. Therefore, (a), (b), (c) and (d,i) naturally stand according to Theorem [S2](#) and the above expression. For (d,ii), we just

prove it for the population version

$$\begin{aligned}
eMD_{pr}^K(G; W) &= \sum_{i=1}^N \sum_{j=1}^N w_{ij} pr\{G \in EM_{H_{i,j}}(\mathbf{G}_1, \mathbf{G}_2, \dots, \mathbf{G}_k)\} \\
&= \sum_{\pi(i)=1}^N \sum_{\pi(j)=1}^N w_{\pi(i)\pi(j)} pr\left\{\pi^G(G) \in EM_{H_{\pi(i),\pi(j)}}(\mathbf{G}_1, \mathbf{G}_2, \dots, \mathbf{G}_k)\right\} \\
&= \sum_{\pi(i)=1}^N \sum_{\pi(j)=1}^N w_{\pi(i)\pi(j)} pr \circ (\pi^G)^{-1}\left\{\pi^G(G) \in EM_{H_{i,j}}(\mathbf{G}_1, \mathbf{G}_2, \dots, \mathbf{G}_k)\right\} \\
&= eMD_{pr \circ (\pi^G)^{-1}}^K(\pi^G(G); \pi^W(W)).
\end{aligned}$$

□

Proof of Theorem 2. For (a), note that

$$\sup_{G \in \mathcal{G}_{[N]}} |eMD_n^K(G; W) - eMD_{pr}^K(G; W)| \leq \sum_{i \neq j} w_{ij} |gMD_n^{H_{i,j},K}(G |_{H_{i,j}}) - gMD_{pr}^{H_{i,j},K}(G |_{H_{i,j}})|.$$

Therefore, the conclusion is implied by the properties of gMD in Theorem S3.

For (b), we just need to transfer the concentration from the whole network to each edge.

Then the conclusion holds because of the properties of gMD in Theorem S3.

The proof of (c) is similar.

□

SVI.3 Proof of Proposition 1

Before prove Proposition 1, we first give a lemma and prove it.

Lemma S1. *A U statistic of order r with kernel h is also a U statistic of order R with kernel h* for all r ≤ R. If h is bounded by M > 0, then h* is also bounded by M.*

Proof of Lemma S1. Assume that U is a U statistic of order r with kernel h(x₁, x₂, ..., x_r),

then

$$\begin{aligned}
U &= \frac{1}{\binom{n}{r}} \sum_{1 \leq i_1 < i_2 < \dots < i_r \leq n} h(X_{i_1}, X_{i_2}, \dots, X_{i_r}) \\
&= \frac{1}{\binom{n}{R}} \frac{\binom{n}{R}}{\binom{n}{r}} \sum_{1 \leq i_1 < i_2 < \dots < i_r \leq n} h(X_{i_1}, X_{i_2}, \dots, X_{i_r}) \\
&= \frac{1}{\binom{n}{R}} \frac{r!(n-r)!}{R!(n-R)!} \sum_{1 \leq i_1 < i_2 < \dots < i_r \leq n} h(X_{i_1}, X_{i_2}, \dots, X_{i_r}) \\
&= \frac{1}{\binom{n}{R}} \sum_{1 \leq i_1 < i_2 < \dots < i_R \leq n} \frac{r!(n-r)!}{R!(n-R)!} \frac{1}{r! \binom{r}{r} \binom{n-r}{R-r}} \sum_{\substack{1 \leq j_1, j_2, \dots, j_r \leq R \\ j_k \neq j_l, k \neq l}} h(X_{i_{j_1}}, X_{i_{j_2}}, \dots, X_{i_{j_r}}) \\
&= \frac{1}{\binom{n}{R}} \sum_{1 \leq i_1 < i_2 < \dots < i_R \leq n} \frac{(R-r)!}{R!} \sum_{\substack{1 \leq j_1, j_2, \dots, j_r \leq R \\ j_k \neq j_l, k \neq l}} h(X_{i_{j_1}}, X_{i_{j_2}}, \dots, X_{i_{j_r}}) \\
&\triangleq \frac{1}{\binom{n}{R}} \sum_{1 \leq i_1 < i_2 < \dots < i_R \leq n} h^*(X_{i_1}, X_{i_2}, \dots, X_{i_R})
\end{aligned}$$

Here, h^* is a symmetric version of h in terms of R elements with a constant correction.

Suppose h is bounded by $M > 0$, i.e. $|h| \leq M$, then

$$\begin{aligned}
|h^*(X_{i_1}, X_{i_2}, \dots, X_{i_R})| &= \left| \frac{(R-r)!}{R!} \sum_{\substack{1 \leq j_1, j_2, \dots, j_r \leq R \\ j_k \neq j_l, k \neq l}} h(X_{i_{j_1}}, X_{i_{j_2}}, \dots, X_{i_{j_r}}) \right| \\
&\leq \frac{(R-r)!}{R!} \sum_{\substack{1 \leq j_1, j_2, \dots, j_r \leq R \\ j_k \neq j_l, k \neq l}} |h(X_{i_{j_1}}, X_{i_{j_2}}, \dots, X_{i_{j_r}})| \\
&= M,
\end{aligned}$$

which suggests h^* is also bounded by M . □

Proof of Proposition 1. Fix $G \in \mathcal{G}_{[N]}$, recall that the sample version $MD_n^k(\cdot)$ is

$$MD_n^k(G) = \binom{n}{k}^{-1} \sum_{1 \leq i_1 < i_2 < \dots < i_k \leq n} \mathbf{1}\{G \in M(\mathbf{G}_{i_1}, \mathbf{G}_{i_2}, \dots, \mathbf{G}_{i_k})\}.$$

It is a U statistics with kernel $h^k(G_1, G_2, \dots, G_k) = \mathbf{1}\{G \in M(G_1, G_2, \dots, G_k)\}$ of order k . Then, we have

- The multiplex is irrelevant to the order of networks and hence the permutation symmetry immediately stands;
- $E(h^k(\mathbf{G}_1, \mathbf{G}_2, \dots, \mathbf{G}_k)) = pr\{G \in M(\mathbf{G}_1, \mathbf{G}_2, \dots, \mathbf{G}_k)\}.$

According to Lemma S1, U statistic of order k is also U statistic of order K for all $k \leq K$ and the fact that the finite summation of U statistics of order K is a U statistic of order K , $gMD_n^K(G)$ is a also U statistic of order K .

Moreover, the kernel of $MD_n^k(G)$ is bounded by 1. Again, with Lemma S1 and the property finite summation of U statistic, the kernel of $gMD_n^K(G)$ is bounded by $K - 1$. Consequently, the second moment of $gMD_n^K(G)$ is finite, i.e., $\mathbb{E}(gMD_n^K(G))^2 < \infty$. Therefore, $gMD_n^K(G) \xrightarrow{a.s.} \mathbb{E}(gMD_n^K(G)) = gMD_{pr}^K(G)$ as $n \rightarrow \infty$ according to Theorem 12.3 in van der Vaart (2000). $\square$

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